

Reflective Journal – Lab 8: Visual Language Models (VLMs)

Overview

Lab 8 was my first real dive into Visual Language Models, systems that combine vision and language. Unlike earlier labs that focused only on images, this one showed me how models can interpret both pictures and text together. I worked with CLIP and BLIP, experimenting with zero-shot classification, captioning, and visual question answering. The point wasn't just to run the code, but to understand how embeddings, generative outputs, and evaluation metrics connect vision and language in practice.

Key Learnings

- **Shared Embedding Spaces:** CLIP taught me how text and image features can be aligned in the same space, which makes zero-shot classification and search possible.
- **Generative Outputs:** BLIP showed me how captions and answers can be generated directly from images, which felt more flexible but also more prone to errors.
- **Evaluation Metrics:** I learned why Recall@K works for retrieval tasks and BLEU works for captioning, and why no single metric captures everything.
- **Fine-Tuning:** Freezing and unfreezing layers made me think about trade-offs between efficiency and adaptability.
- **Limitations:** I saw how vague prompts or ambiguous images can lead to hallucinations or biased outputs.

Application to Practice

What struck me is how these same ideas apply in real industries. In retail, VLMs can power product search by connecting customer queries to catalog images. In healthcare, they could help generate descriptive notes from medical scans, making information easier to share. In education, captioning and VQA can make visual materials more accessible to students. Even in creative industries like design or media, VLMs can speed up brainstorming by linking text prompts to visual references. The lab made me realize that multimodal AI isn't just a research concept — it's already shaping how information is searched, explained, and communicated across fields.

Reflection

This lab pushed me to think beyond single-modality AI. I realized that combining vision and language unlocks a new level of usability: instead of just detecting or classifying, the model can explain and communicate. At the same time, I saw how fragile these systems can be. A poorly phrased prompt or an ambiguous image can lead to misleading outputs, which means human judgment is still essential. For me, the biggest takeaway is balance — VLMs are powerful tools, but they need careful evaluation and context-aware use. Lab 8 gave me confidence in experimenting with multimodal systems and reminded me that the future of AI is not just about seeing or speaking, but about connecting the two in ways that are useful and trustworthy.